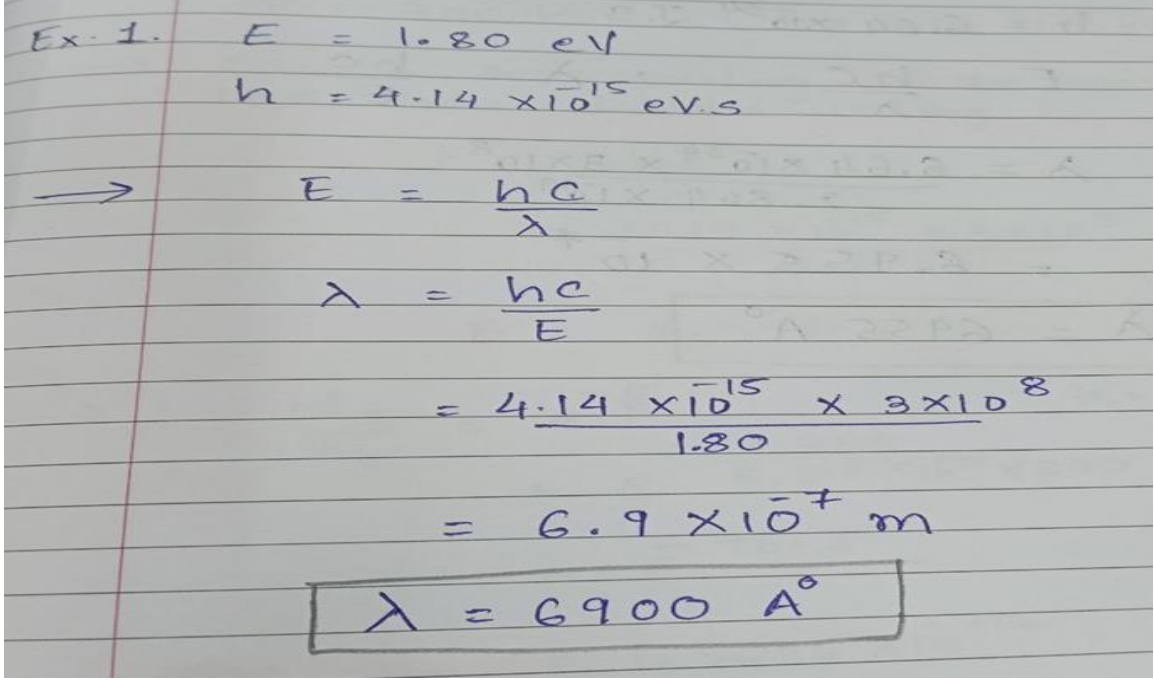
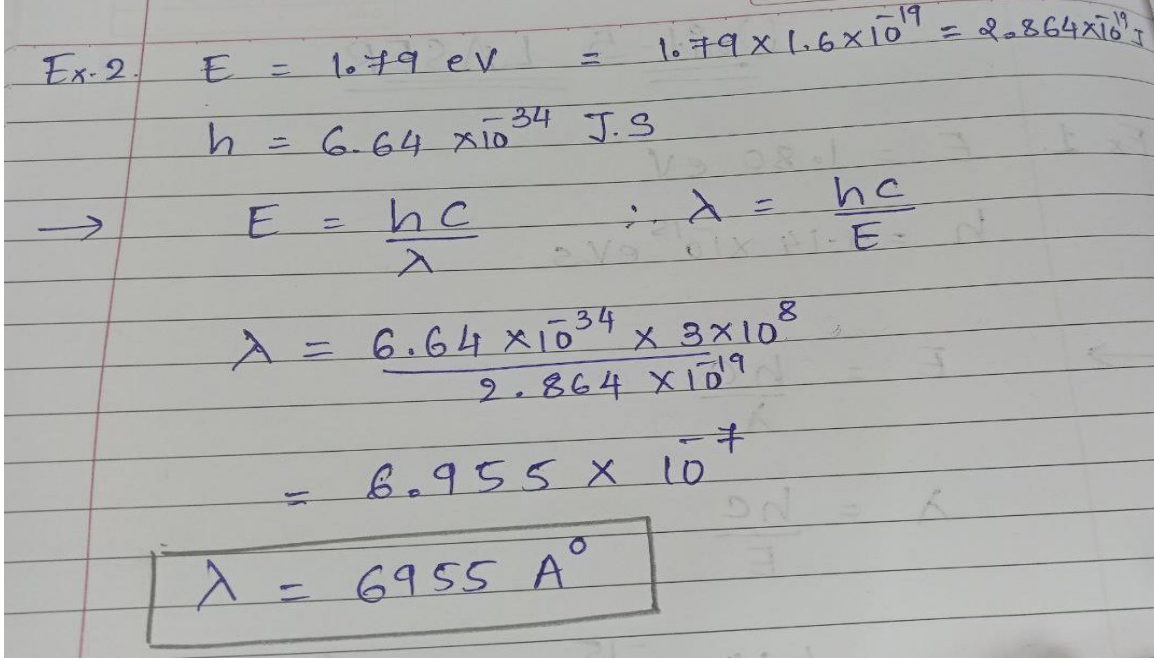
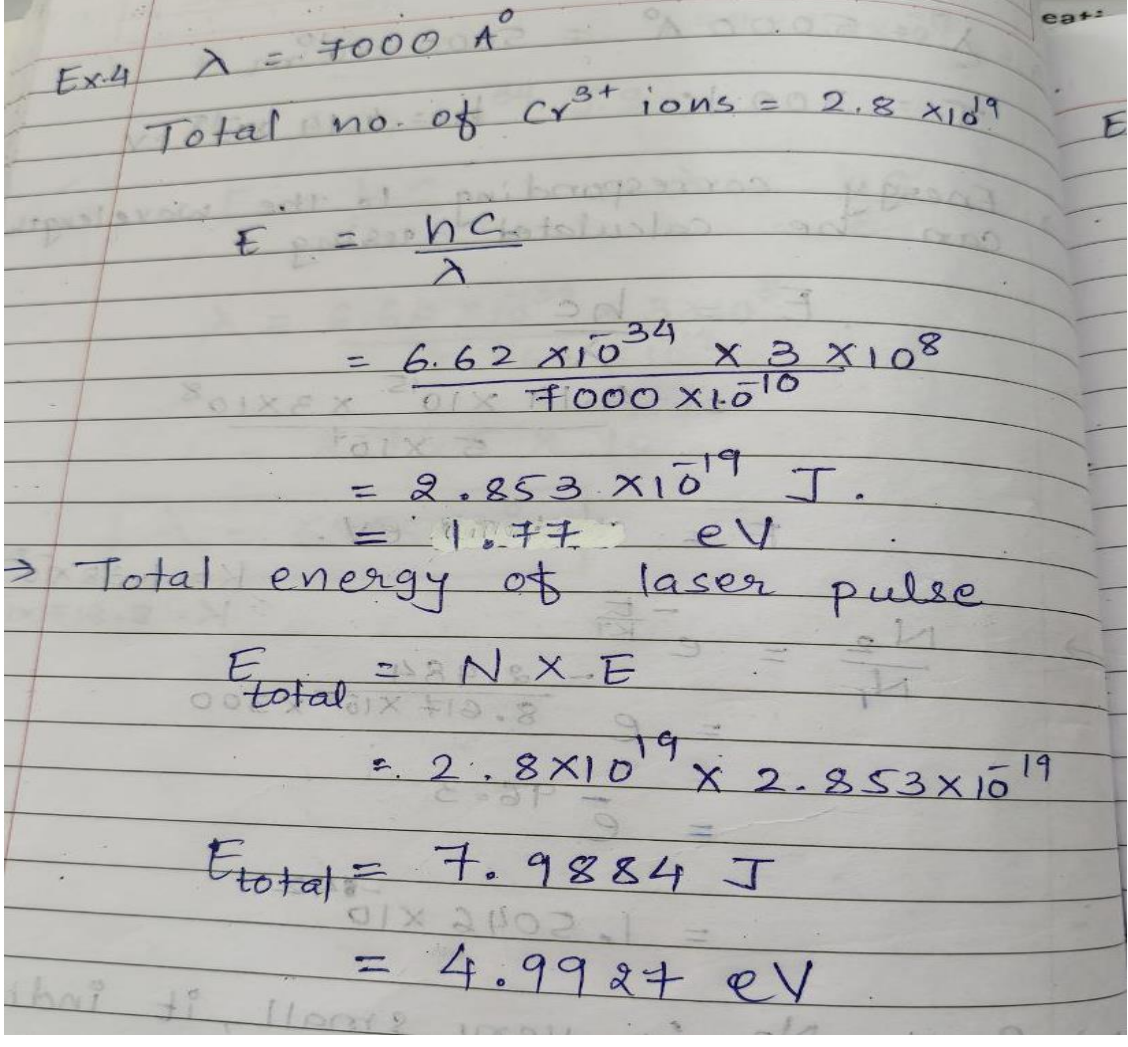


Examples

1.	Calculate the wavelength of Laser light if the separation between metastable state and lower energy state is 1.80 eV. (Consider Planck's constant is 4.14×10^{-15} eV.s) (Jun-2019) [NLJIET]	3
	 Ex-1. $E = 1.80 \text{ eV}$ $h = 4.14 \times 10^{-15} \text{ eV.s}$ → $E = \frac{hc}{\lambda}$ $\lambda = \frac{hc}{E}$ $= \frac{4.14 \times 10^{-15} \times 3 \times 10^8}{1.80}$ $= 6.9 \times 10^{-7} \text{ m}$ $\lambda = 6900 \text{ \AA}$	
2.	What is the wavelength of light of Ruby Laser if the separation between metastable state and lower energy state is 1.79 eV. Given that Planck's constant = 6.64×10^{-34} Js. (Jan-2020) [NLJIET]	3
	 Ex-2. $E = 1.79 \text{ eV} = 1.79 \times 1.6 \times 10^{-19} = 2.864 \times 10^{-19} \text{ J}$ $h = 6.64 \times 10^{-34} \text{ J.s}$ → $E = \frac{hc}{\lambda} \therefore \lambda = \frac{hc}{E}$ $\lambda = \frac{6.64 \times 10^{-34} \times 3 \times 10^8}{2.864 \times 10^{-19}}$ $= 6.955 \times 10^{-7}$ $\lambda = 6955 \text{ \AA}$	

3.	In Ruby laser, total number of Cr^{+3} ions in excited state are 2.8×10^{19} . If the laser emits radiation of wavelength $7000 \text{ \AA}$, calculate energy of laser pulse. (Mar-2023) [NLJIET]	3
	 Ex-4 $\lambda = 7000 \text{ \AA}$ Total no. of Cr^{3+} ions $= 2.8 \times 10^{19}$ $E = \frac{hc}{\lambda}$ $= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{7000 \times 10^{-10}}$ $= 2.853 \times 10^{-19} \text{ J.}$ $= 1.77 \text{ eV}$ → Total energy of laser pulse $E_{\text{total}} = N \times E$ $= 2.8 \times 10^{19} \times 2.853 \times 10^{-19}$ $E_{\text{total}} = 7.9884 \text{ J}$ $= 4.9927 \text{ eV}$	
4.	In He-Ne laser system, the two energy levels of Ne involved in lasing action have energy values of 20.66 eV and 18.70 eV . Population inversion occurs between these two levels. What will be the wavelength of a laser beam produced? What will be the population of the metastable energy level with respect to the upper excited level at 27°C ? (Jan-2024) [NLJIET]	3

Ex-5. $E_1 = 18.70 \text{ eV}$
 $E_2 = 20.66 \text{ eV}$

$T = 27^\circ\text{C} = 273 + 27 = 300 \text{ K}$

$\rightarrow E_2 - E_1 = h\nu = \frac{hc}{\lambda}$

$\therefore \lambda = \frac{hc}{E_2 - E_1}$

$= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{(20.66 - 18.70) \times 1.6 \times 10^{-19}}$

$= 6332.9 \times 10^{-10} \text{ m}$

$\lambda = 6332.9 \text{ \AA}$

$\frac{N_2}{N_1} = e^{-h\nu/kT} = e^{-hc/\lambda kT}$

$= \exp \left[- \frac{(6.62 \times 10^{-34} \times 3 \times 10^8)}{6332.9 \times 10^{-10} \times 1.38 \times 10^{-23} \times 300} \right]$

$= \exp \left[- \frac{19.86 \times 10^{-26}}{2.6218 \times 10^{-24}} \right]$

$\frac{N_2}{N_1} = \exp [-75.74] = 1.27 \times 10^{-33}$

5. In He-Ne gas laser, Ne atoms emit laser photons of wavelength 632.8 nm while electrons make transitions from E_3 state to E_2 state. Find the energy difference between these two states. ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$ and $c = 3 \times 10^8 \text{ m/s}$)

(Jun-2025-New) [NLJIET]

6 $\lambda = 632.8 \text{ nm}$
 $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$
 $c = 3 \times 10^8 \text{ m/s}$
 $E_3 - E_2 = ?$

$E_3 - E_2 = \frac{hc}{\lambda}$

$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{632.8 \times 10^{-9}}$

$= 0.03143 \times 10^{-17}$

$E_3 - E_2 = 3.143 \times 10^{-19} \text{ J}$

$E_3 - E_2 = 1.9643 \text{ eV}$

3

6. A ruby laser emits light of wavelength 694.3 nm. Calculate the frequency and the energy of the laser photons emitted. ($h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$ and $c = 3 \times 10^8 \text{ m/s}$)
(Jun-2025-New) [NLJIET]

3

$\lambda = 694.3 \text{ nm}$
 $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$
 $c = 3 \times 10^8 \text{ m/s}$
 $f = ?$
 $E = ?$

$$f = \frac{c}{\lambda}$$

$$= \frac{3 \times 10^8}{694.3 \times 10^{-9}}$$

$$= 0.004320 \times 10^{17}$$

$$f = 4.32 \times 10^{14} \text{ Hz}$$

$$E = hf$$

$$= 6.63 \times 10^{-34} \times 4.32 \times 10^{14}$$

$$= 28.6416 \times 10^{-20}$$

$$E = 2.8642 \times 10^{-19} \text{ J}$$

$$E = 1.7901 \text{ eV}$$

7. Using Einstein's theory, show that in the optical region, say, at $\lambda=5000 \text{ \AA}$ and $T=300 \text{ K}$, the amplification is not possible.

4

(Mar-2023) [NLJIET]

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Ex. 3. $\lambda = 5000 \text{ \AA} = 5000 \times 10^{-10} \text{ m}$

$T = 300 \text{ K}$ $h = 4.14 \times 10^{-15} \text{ eV}$

→ Energy corresponding to the wavelength can be calculated using

$$E = \frac{hc}{\lambda}$$
$$= \frac{4.14 \times 10^{-15} \times 3 \times 10^8}{5 \times 10^{-7}}$$
$$E = 2.484 \text{ eV.}$$

→ $\frac{N_2}{N_1} = e^{-\frac{E}{kT}}$ $k = 8.617 \times 10^{-5} \text{ eV}$

$$= e^{-\frac{2.484}{8.617 \times 10^{-5} \times 300}}$$
$$= e^{-96.3}$$
$$= 1.5046 \times 10^{-42}$$

→ Ratio $\frac{N_2}{N_1}$ is very small, it indicates No. of atoms in excited state is negligible to lower state.

→ Therefore amplification is not Possible